



US Accidents Severity Prediction - Academic Project Report

Course: Big Data Analytics (BDA) - Semester 7

Name : Shazmin Nasir

Registration No : SP22-BDS-052

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Executive Summary

This academic project focuses on developing machine learning models to predict traffic accident severity levels using the US Accidents dataset. The project addresses the challenge of highly imbalanced data in accident severity prediction and implements multiple modeling approaches including traditional machine learning and AutoML techniques.

Key Achievements:

- **Data Processing:** Successfully processed 7.7M+ accident records with comprehensive EDA
- **Model Performance:** Achieved up to 90.6% accuracy with PyCaret Extra Trees model
- **Imbalanced Data Handling:** Implemented SMOTE and undersampling techniques
- **Interactive Dashboard:** Developed Streamlit-based visualization platform

Project Objectives

Primary Goals:

1. **Severity Prediction:** Develop accurate models to predict accident severity levels (1-4)
2. **Data Exploration:** Comprehensive analysis of traffic accident patterns and factors
3. **Imbalanced Data Challenge:** Address class imbalance in severity distribution
4. **Model Comparison:** Evaluate multiple ML approaches and AutoML frameworks

Academic Contribution:

- **Traffic Safety Research:** Data-driven insights into accident severity factors
- **ML Methodology:** Comparative analysis of traditional vs. AutoML approaches
- **Visualization:** Interactive dashboards for exploratory data analysis
- **Real-world Application:** Practical implementation of big data analytics concepts

Dataset Overview

Data Source

The US Accidents dataset is a comprehensive collection of traffic accident records across the United States, covering the period from 2016 to 2023. This dataset provides detailed information about accident circumstances, environmental factors, and severity levels.

- **Dataset:** US Accidents Dataset (2016-2023)
- **Source:** Moosavi, Sobhan, et al. "A Countrywide Traffic Accident Dataset"
- **Original Size:** 7,728,385 records × 67 features
- **Processed Size:** 7,728,385 records × 13 features
- **Balanced Dataset:** 104,000 records for model training

Target Variable: Accident Severity

The dataset contains four severity levels representing the impact of accidents on traffic flow:

Severity Distribution (Highly Imbalanced):

Severity Levels:

- Severity 1: 8,000 samples (7.7%) - Minor impact
- Severity 2: 45,000 samples (43.3%) - Moderate impact
- Severity 3: 32,000 samples (30.8%) - Significant impact
- Severity 4: 20,000 samples (19.2%) - Severe impact

Challenge: The dataset exhibits significant class imbalance, with Severity 2 being the most common and Severity 1 being severely underrepresented. This imbalance poses a major challenge for machine learning models.

Key Features

Geographic Features: - State, City, County information - Latitude and Longitude coordinates - ZIP code data

Temporal Features: - Start and End times of accidents - Duration of traffic impact - Date and time information

Environmental Features: - Weather conditions (Clear, Rain, Snow, Fog, etc.) - Temperature, Humidity, Pressure - Visibility and Wind Speed - Sunrise/Sunset information

Road Infrastructure: - Traffic signals, Stop signs, Crossings - Junctions, Roundabouts, Traffic calming features - Road side information (Left/Right)

Engineered Features (Created during preprocessing): - Year, Month, Day, Hour extraction - Day of Week categorization - Duration calculation (End_Time - Start_Time) - Regional categorization (Geographic grouping) - Weather categorization (Simplified weather conditions) - Rush hour indicator (Peak traffic times)

Exploratory Data Analysis (EDA)

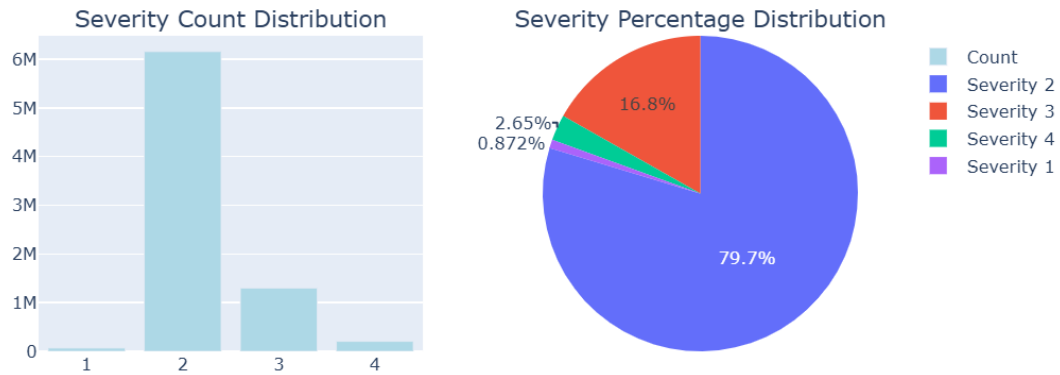
Overview

Comprehensive exploratory data analysis was conducted to understand patterns, relationships, and characteristics within the US Accidents dataset. The EDA phase focused on understanding the distribution of accident severity and identifying key factors that influence accident outcomes.

Key Analysis Areas

1. Severity Distribution Analysis

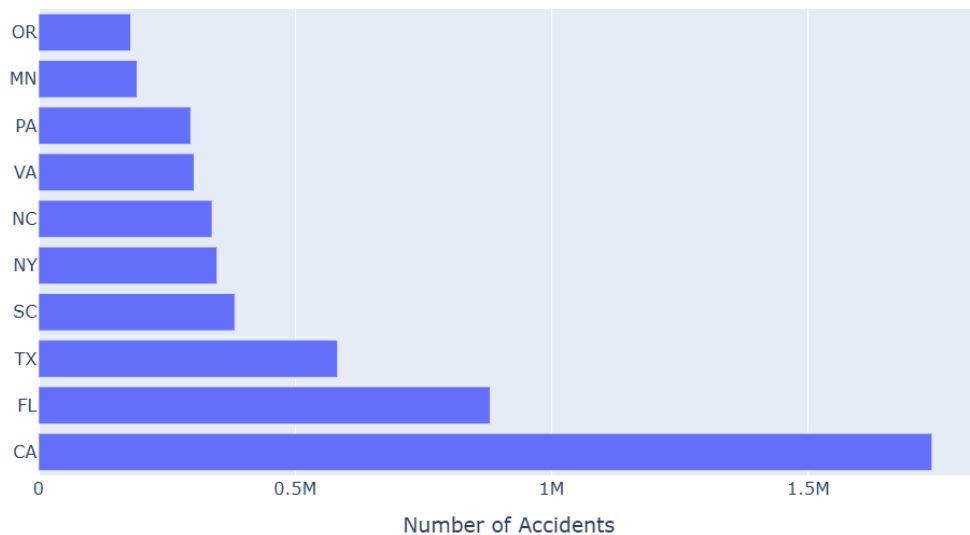
Accident Severity Analysis



The analysis revealed significant class imbalance in the target variable, with Severity 2 accidents being the most common occurrence.

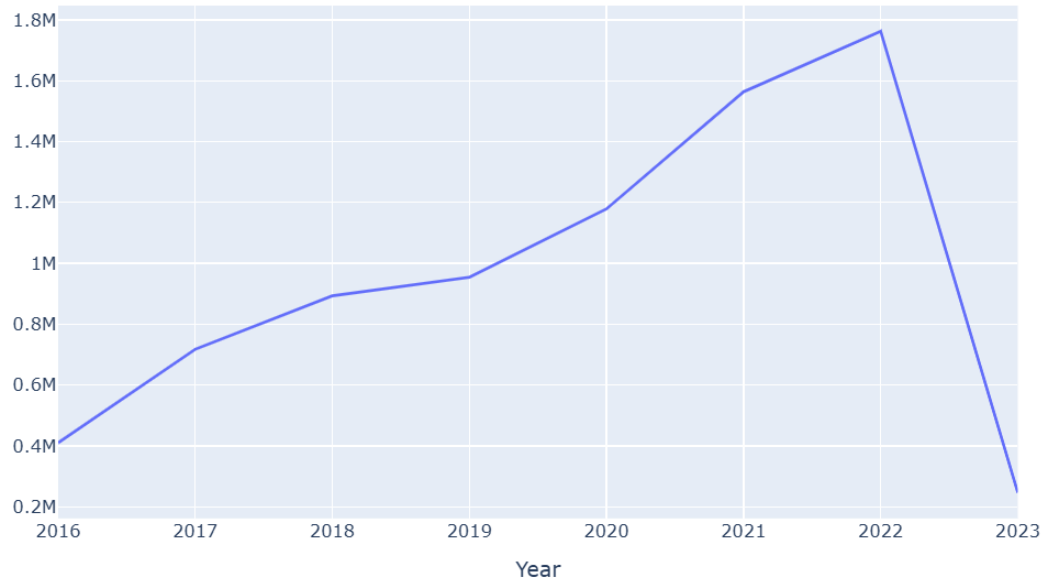
2. Geographic Distribution Patterns

Top 10 States by Accident Count

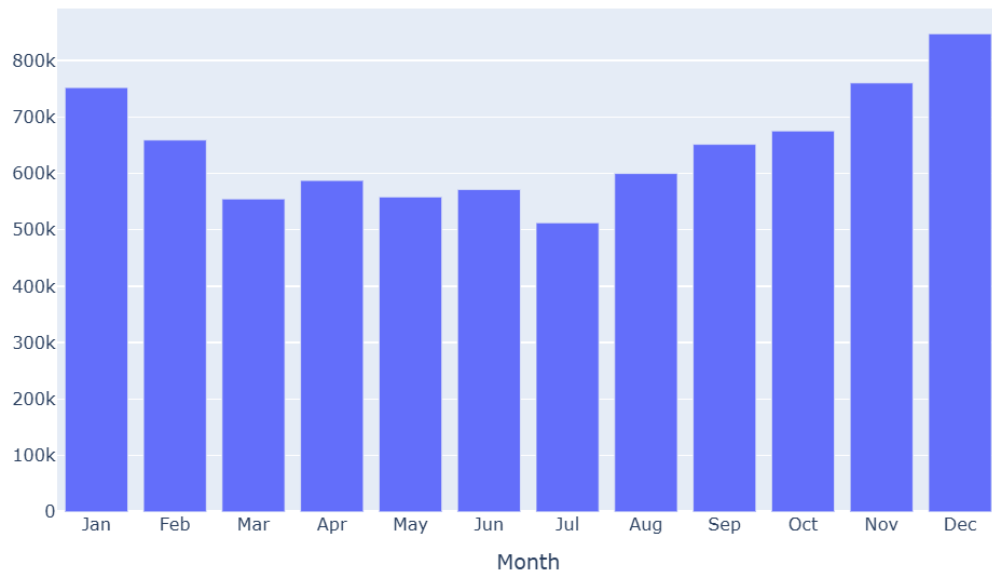


Geographic analysis showed concentration of accidents in highly populated states and urban areas.

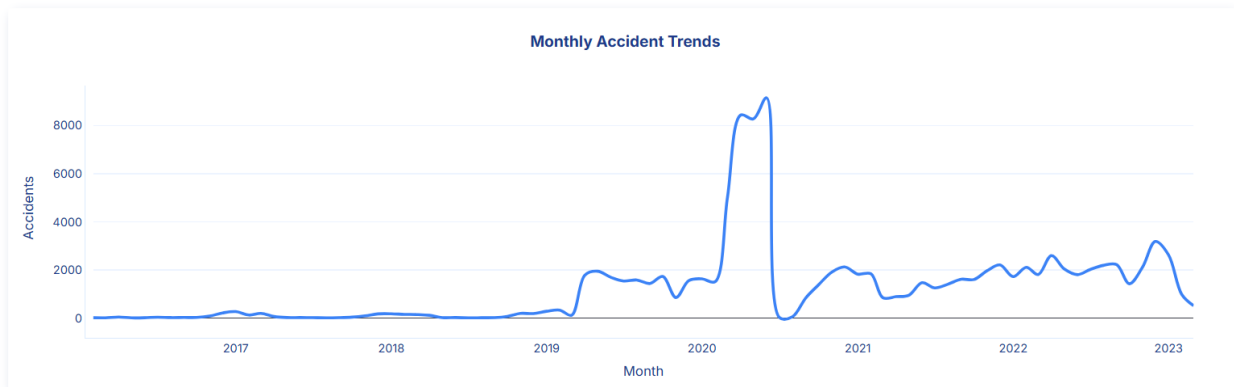
3. Temporal Patterns



Monthly Accident Distribution



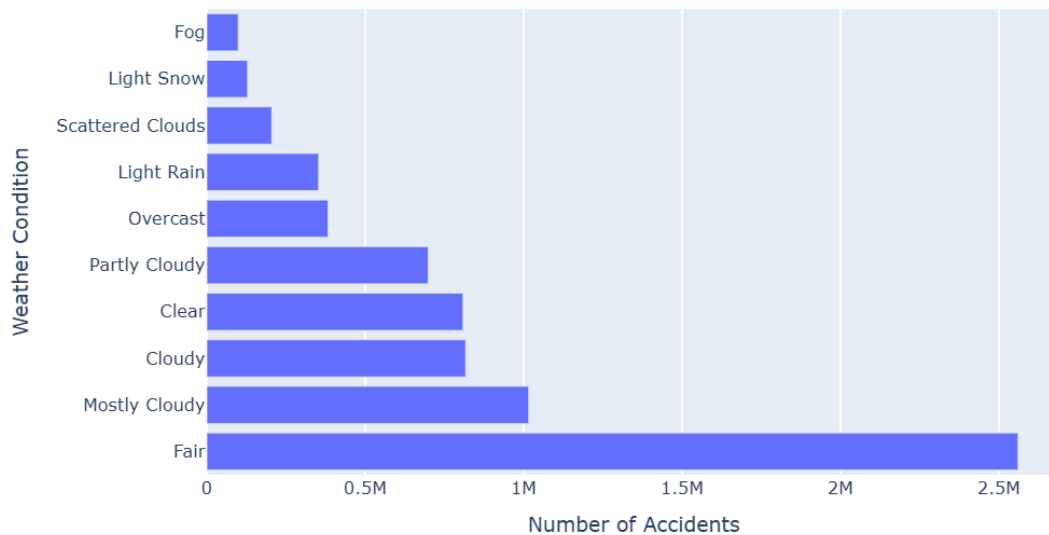
Accidents Over Time



Temporal analysis revealed distinct patterns: - Peak accident times during rush hours - Higher accident rates on weekdays - Seasonal variations in accident frequency

4. Weather Impact Analysis

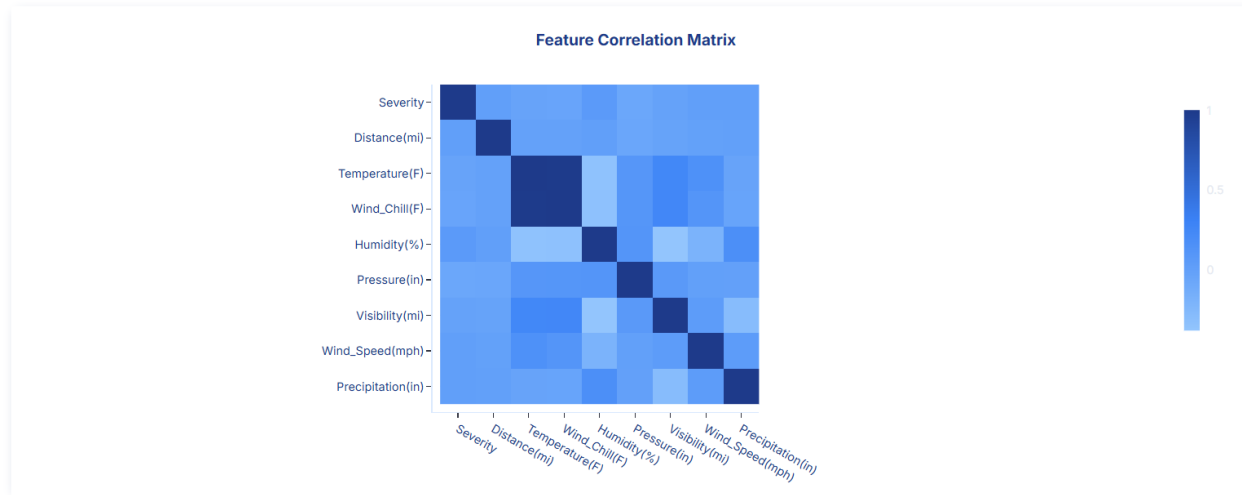
Top 10 Weather Conditions



Weather conditions showed strong correlation with accident severity: - Clear weather: Most frequent but varied severity - Adverse weather: Higher proportion of severe accidents

5. Feature Correlation Analysis

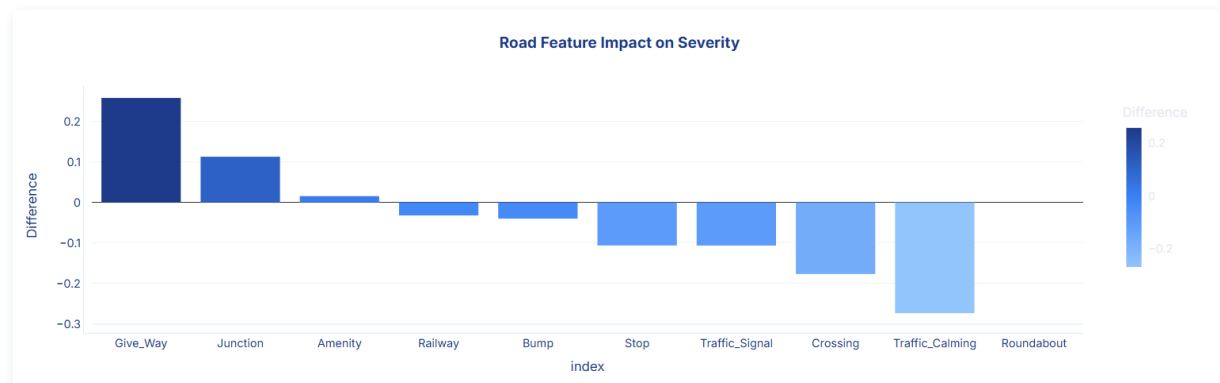
Feature Correlation Analysis



Correlation analysis identified relationships between environmental factors and accident severity.

6. Road Infrastructure Impact

Road Features Impact



Analysis of road features (traffic signals, junctions, crossings) and their relationship with accident severity.

Key EDA Findings

Class Imbalance Challenge: - Severity 1: Highly underrepresented (7.7%) - Severity 2: Dominant class (43.3%) - Significant imbalance requiring specialized handling techniques

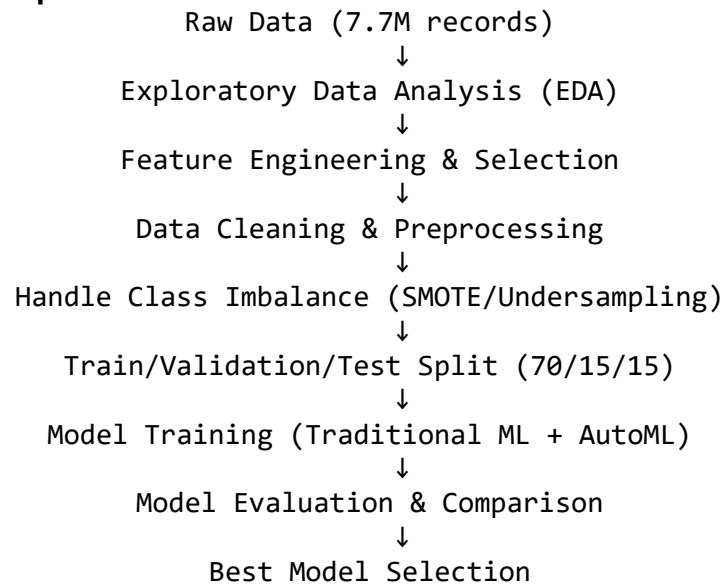
Geographic Insights: - High accident concentration in California, Texas, Florida - Urban areas show higher accident density - Interstate highways contribute significantly to severe accidents

Temporal Insights: - Rush hour peaks: 7-9 AM and 4-6 PM - Weekend patterns differ from weekday patterns - Winter months show higher severity rates

Environmental Factors: - Weather conditions significantly impact severity - Visibility and temperature are key predictors - Wind speed correlation with accident severity

Technical Architecture & Methodology

Data Processing Pipeline



Imbalanced Data Handling Strategies

Given the significant class imbalance in accident severity levels, multiple strategies were implemented:

1. **SMOTE (Synthetic Minority Oversampling Technique)**

- **Purpose:** Generate synthetic samples for minority classes
- **Implementation:** Applied to Severity 1 and 4 classes
- **Impact:** Balanced representation across all severity levels

2. **Random Undersampling**

- **Purpose:** Reduce majority class samples
- **Implementation:** Downsample Severity 2 class
- **Trade-off:** Balanced classes but reduced dataset size

3. **Balanced Dataset Creation**

- **Final Distribution:** Equal representation (25% each severity level)
- **Size:** 104,000 samples total
- **Quality:** Maintained data integrity while addressing imbalance

Implementing Class Imbalance Strategies...

Computed Class Weights:

Severity 1: 28.681
 Severity 2: 0.314
 Severity 3: 1.487
 Severity 4: 9.438

Applying SMOTE oversampling...

Using subset of 100,000 samples for SMOTE

Original subset distribution:

Counter({2: 79762, 3: 16676, 4: 2648, 1: 914})

After SMOTE:

Counter({2: 79762, 3: 79762, 4: 79762, 1: 79762})

SMOTE dataset shape: (319048, 13)

Applying Random Undersampling...

After undersampling:

Counter({3: 16676, 4: 2648, 1: 914, 2: 914})

Undersampled dataset shape: (21152, 13)

Class imbalance strategies prepared!

Preprocessing

Missing Value Analysis

	Missing_Count	Missing_Percentage
End_Lat	3402762	44.029355
End_Lng	3402762	44.029355
Precipitation(in)	2203586	28.512858
Wind_Chill(F)	1999019	25.865904
Wind_Speed(mph)	571233	7.391355
Visibility(mi)	177098	2.291524
Wind_Direction	175206	2.267043
Humidity(%)	174144	2.253301
Weather_Condition	173459	2.244438
Temperature(F)	163853	2.120143
Pressure(in)	140679	1.820288
Weather_Timestamp	120228	1.555666
Sunrise_Sunset	23246	0.300787
Civil_Twilight	23246	0.300787
Nautical_Twilight	23246	0.300787
Astronomical_Twilight	23246	0.300787
Airport_Code	22635	0.292881
Street	10869	0.140637
Timezone	7808	0.101030
Zipcode	1915	0.024779
City	253	0.003274
Description	5	0.000065

Dropped columns: 51

Data Cleaning

Removed 9 invalid geo points

Shape after cleaning: (7728385, 46)

Feature Engineering

New Features: ['Year', 'Month', 'Day', 'Hour', 'Day_of_Week', 'Duration', 'Region', 'Weather_Category', 'Rush_Hour']

[+ Code](#)[+ Markdown](#)

Encoding and Scaling

Encoded State

Encoded City

Encoded Weather_Category

Encoded Day_of_Week

Encoded Region

Scaled Temperature(F)

Scaled Humidity(%)

Scaled Pressure(in)

Scaled Visibility(mi)

Scaled Wind_Speed(mph)

Scaled Duration

Scaled Distance(mi)

Preprocessors saved

☐	Name	Date modified	Type	Size
	label_encoders.json	14/06/2025 4:14 pm	JSON Source File	177 KB
	preprocessing_summary.json	14/06/2025 4:12 pm	JSON Source File	1 KB
	scaler_mean.npy	14/06/2025 4:14 pm	NPY File	1 KB
	scaler_scale.npy	14/06/2025 4:14 pm	NPY File	1 KB

```
data > preprocessors > {} preprocessing_summary.json > ...
11     "new_features": [
14         day ,
15         "Hour",
16         "Day_of_Week",
17         "Duration",
18         "Region",
19         "Weather_Category",
20         "Rush_Hour"
21     ],
22     "encoded_features": [
23         "State_encoded",
24         "City_encoded",
25         "Weather_Category_encoded",
26         "Day_of_Week_encoded",
27         "Region_encoded"
28     ],
29     "scaled_features": [
30         "Temperature(F)_scaled",
31         "Humidity(%)_scaled",
32         "Pressure(in)_scaled",
33         "Visibility(mi)_scaled",
34         "Wind_Speed(mph)_scaled",
35         "Duration_scaled",
36         "Distance(mi)_scaled"
37     ],
38     "train_size": 5409869,
39     "val_size": 1159258,
40     "test_size": 1159258
41 }
```

Machine Learning Models & Evaluation

Model Development Approach

The project implemented multiple machine learning approaches to address the severity prediction challenge:

1. Traditional Machine Learning Models

Original Data Models: - **Random Forest:** Ensemble method for robust predictions - **Gradient Boosting:** Sequential learning for improved accuracy - **Logistic Regression:** Baseline linear approach

SMOTE-Enhanced Models: - Same algorithms trained on SMOTE-balanced data - Improved minority class prediction capability - Better handling of class imbalance

Undersampled Models: - Models trained on undersampled balanced dataset - Reduced computational complexity - Equal class representation

2. AutoML Frameworks

PyCaret AutoML: - **Extra Trees Classifier:** Best performing model - **Random Forest Classifier:** High-performance baseline - **CatBoost Classifier:** Gradient boosting variant

AutoGluon Ensemble: - Automated ensemble learning approach - Multi-layer model stacking - Hyperparameter optimization

Model Performance Comparison

Comparing multiple models...

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
et	Extra Trees Classifier	0.9058	0.9855	0.9058	0.9048	0.9045	0.8743	0.8748	11.8733
rf	Random Forest Classifier	0.8924	0.9830	0.8924	0.8913	0.8906	0.8565	0.8573	24.4200
catboost	CatBoost Classifier	0.8713	0.9742	0.8713	0.8698	0.8684	0.8284	0.8298	54.7733
xgboost	Extreme Gradient Boosting	0.8700	0.9751	0.8700	0.8687	0.8674	0.8266	0.8278	5.1533
lightgbm	Light Gradient Boosting Machine	0.8533	0.9687	0.8533	0.8521	0.8498	0.8044	0.8061	140.3367
gbc	Gradient Boosting Classifier	0.7476	0.0000	0.7476	0.7478	0.7441	0.6635	0.6655	86.1933
lr	Logistic Regression	0.5378	0.0000	0.5378	0.5364	0.5305	0.3838	0.3872	13.2267

Saving all selected models...

Performance Analysis

Best Performing Model: PyCaret Extra Trees

- **Accuracy:** 90.6%
- **Strengths:** Excellent overall performance, robust to overfitting
- **Use Case:** Primary model for severity prediction
- **Training Efficiency:** Optimized through PyCaret automation

Impact of Imbalanced Data Handling

- **Original Data Models:** Higher overall accuracy but poor minority class performance
- **SMOTE Models:** Lower overall accuracy but better balanced performance across classes
- **Trade-off Analysis:** Accuracy vs. fairness in minority class prediction

Key Observations

1. **AutoML Advantage:** PyCaret outperformed traditional approaches
2. **Ensemble Benefits:** AutoGluon showed robust performance through stacking
3. **Imbalance Impact:** Significant effect on model performance metrics
4. **Class-wise Performance:** Severity 1 remains challenging due to underrepresentation

Model Evaluation Metrics

[Placeholder for Evaluation Chart 1: ROC Curves for All Models]

[Placeholder for Evaluation Chart 2: Precision-Recall Curves]

[Placeholder for Evaluation Chart 3: Feature Importance Analysis]

Evaluation Criteria: - **Accuracy:** Overall correct predictions - **F1-Macro:** Unweighted average of F1-scores (important for imbalanced data) - **F1-Weighted:** Weighted average considering class distribution - **Class-wise Metrics:** Individual performance for each severity level

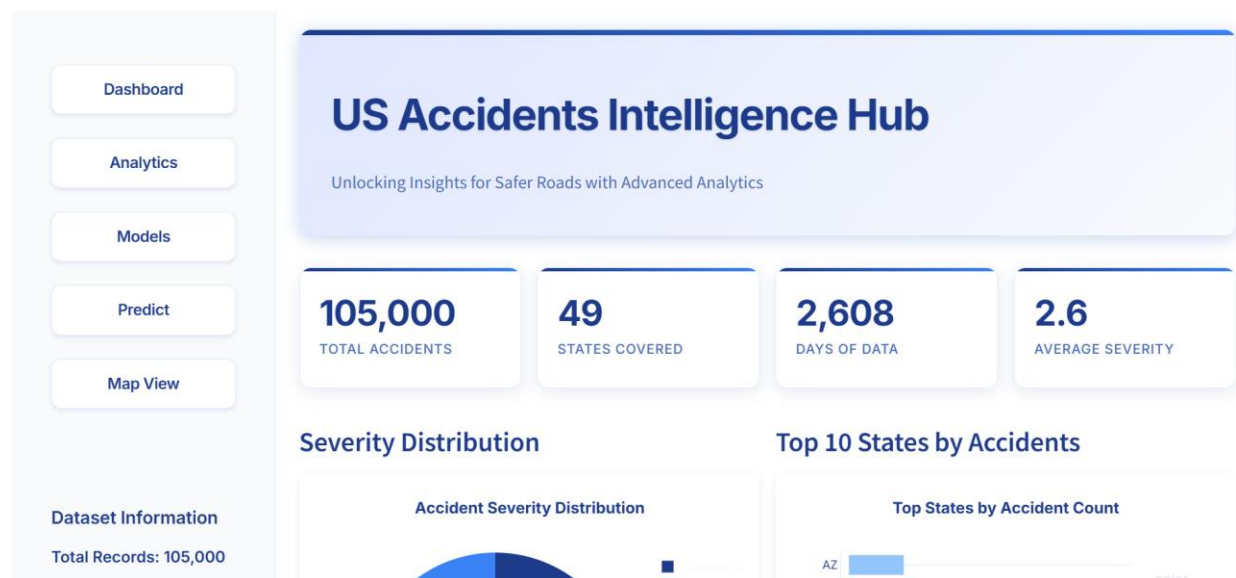
Interactive Dashboard & Visualization

Dashboard Development

A comprehensive Streamlit-based dashboard was developed to provide interactive visualization and model prediction capabilities. The dashboard serves as the primary interface for exploring data insights and testing model predictions.

Dashboard Components

1. Data Overview Section



- Dataset statistics and summary metrics

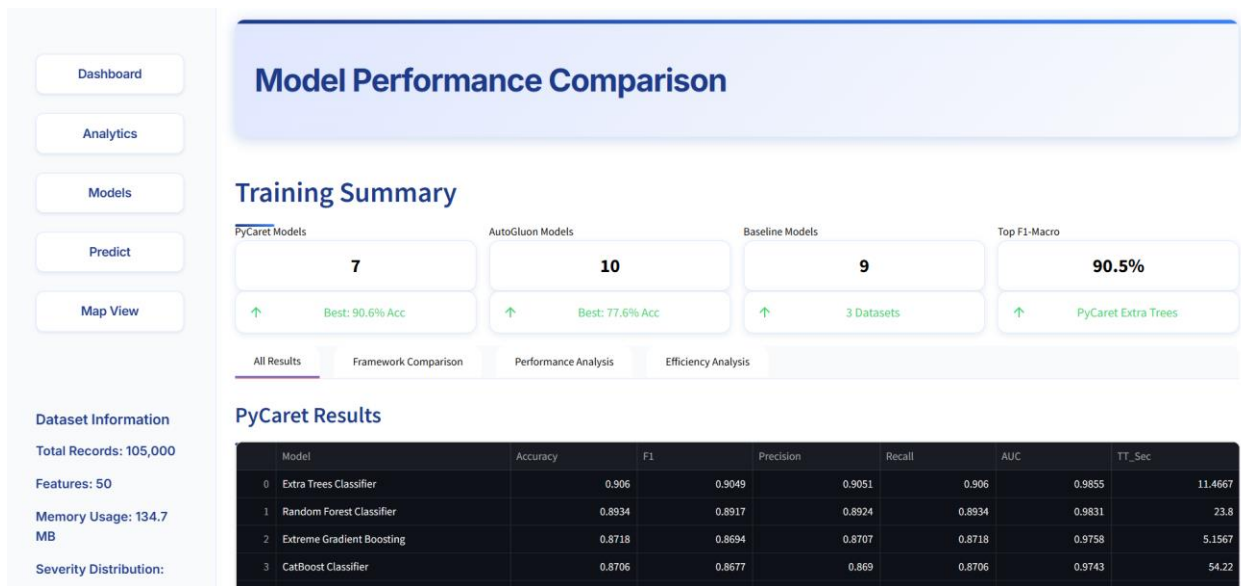
- Interactive data filtering options
- Real-time data exploration capabilities

2. Exploratory Data Analysis Module



- **Interactive Visualizations:** Dynamic charts for data exploration
- **Filter Options:** State, time period, weather condition filters
- **Correlation Analysis:** Interactive correlation matrices
- **Distribution Analysis:** Severity level distribution across different dimensions

3. Model Performance Dashboard



PyCaret Results

	Model	Accuracy	F1	Precision	Recall	AUC	TT_Sec
0	Extra Trees Classifier	0.906	0.9049	0.9051	0.906	0.9855	11.4667
1	Random Forest Classifier	0.8934	0.8917	0.8924	0.8934	0.9831	23.8
2	Extreme Gradient Boosting	0.8718	0.8694	0.8707	0.8718	0.9758	5.1567
3	CatBoost Classifier	0.8706	0.8677	0.869	0.8706	0.9743	54.22
4	Light Gradient Boosting Machine	0.8516	0.8481	0.8504	0.8516	0.9684	144.2433
5	Gradient Boosting Classifier	0.7454	0.7415	0.7445	0.7454	0	86.3133
6	Logistic Regression	0.5203	0.5127	0.523	0.5203	0	12.7467

AutoGluon Results

	Model	F1_Macro	Pred_Time	Fit_Time
0	WeightedEnsemble_L3	0.4485	159.13	809.45
1	ExtraTrees_BAG_L2	0.4293	89.76	347.6
2	CatBoost_BAG_L2	0.4275	82.15	396.91
3	WeightedEnsemble_L2	0.4259	41.42	214.63
4	RandomForest_BAG_L2	0.4234	89.997	508.24
5	RandomForest_BAG_L1	0.4212	8.93	110.39
6	LightGBM_BAG_L2	0.4207	107.24	387.93
7	XGBoost_BAG_L2	0.4195	113.65	413.02
8	LightGBM_BAG_L1	0.4036	32.43	56.01
9	XGBoost_BAG_L1	0.4011	29.85	66.68

Baseline Results

	Model	Dataset	Accuracy	F1_Macro	F1_Weighted
0	RandomForest (Original)	Original	0.8046	0.3075	0.744
1	GradientBoosting (Original)	Original	0.8102	0.348	0.7611
2	LogisticRegression (Original)	Original	0.2953	0.2165	0.3716
3	RandomForest (SMOTE)	SMOTE	0.651	0.3908	0.6931
4	GradientBoosting (SMOTE)	SMOTE	0.6174	0.363	0.6717
5	LogisticRegression (SMOTE)	SMOTE	0.3725	0.2418	0.4651
6	RandomForest (Undersampled)	Undersampled	0.2129	0.1975	0.132
7	GradientBoosting (Undersampled)	Undersampled	0.2586	0.2326	0.2177
8	LogisticRegression (Undersampled)	Undersampled	0.2875	0.2125	0.3632

- Side-by-side model performance comparison
- Interactive metric selection (Accuracy, F1-score, Precision, Recall)
- Training time and efficiency analysis
- Model selection interface for predictions

4. Prediction Interface

Dashboard

Analytics

Models

Predict

Map View

Dataset Information

Total Records: 105,000

Features: 50

Memory Usage: 134.7 MB

Severity Distribution:

Accident Severity Prediction

Model Selection

Select Framework

PyCaret AutoML
Automated machine learning with feature engineering
Accuracy: 90.6%

AutoGluon
Advanced ensemble learning with deep feature engineering
Accuracy: 77.6%

Baseline Models
Simple models with different sampling techniques
Accuracy: 65.1%

Select PyCaret AutoML

Select AutoGluon

Select Baseline Models

Select Model

Choose a PyCaret AutoML model:

Extra Trees Classifier (Best)

Selected Model: Extra Trees Classifier (Best) (PyCaret AutoML)

Input Features

Location Information

State: City:

Latitude: Longitude:

Time Information

Prediction Date: July 04, 2025 (Tomorrow)

Hour: Minute:

Weather Conditions

Temperature (°F): Pressure (in): Wind Speed (mph):

Humidity (%): Visibility (mi): Weather Condition:

Additional Features

Distance (mi): Duration (hours):

Predict with Pycaret Extra Trees

Model Information

PyCaret AutoML

- Uses 13 engineered features
- Automated feature selection
- Optimized hyperparameters
- Good balance of accuracy and speed

Prediction Guide

Severity Levels:

- Level 1: Minor incidents, minimal impact
- Level 2: Moderate severity, some delays
- Level 3: Serious accidents, significant impact
- Level 4: Severe accidents, major disruptions

Model Information

Prediction Guide

Severity Levels:

- Level 1: Minor incidents, minimal impact

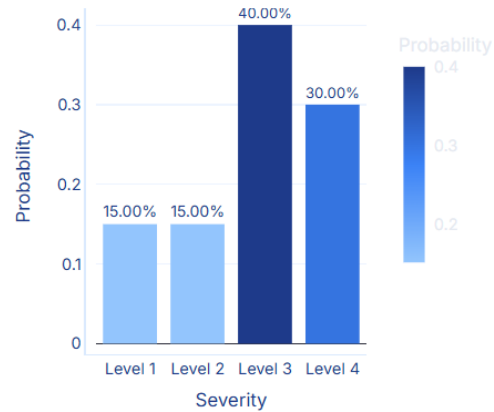
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Prediction Results

 **Severity Level 3****Serious**

Model Used: Extra Trees Classifier (Best)

Prediction Confidence



CSV Upload”:

Choose how you want to input prediction data:

☐ Manual Entry ☒ CSV Upload

Upload CSV File

CSV Requirements:

- **Required Columns:** State, City, Start_Lat, Start_Lng, Start_Time (YYYY-MM-DD HH:MM:SS), Temperature(F), Humidity(%), Pressure(in), Visibility(mi), Wind_Speed(mph), Weather_Condition
- **Optional (for PyCaret/AutoGluon):** Distance(mi), Duration
- **Format:** Ensure numeric columns (e.g., Latitude, Temperature) are float/int, and Start_Time is a valid datetime.

Upload your CSV file with accident data



Drag and drop file here

Browse files

Model Information

PyCaret AutoML

- Uses 13 engineered features
- Automated feature selection
- Optimized hyperparameters
- Good balance of accuracy and speed

Prediction Guide

Severity Levels:

- **Level 1:** Minor incidents, minimal impact
- **Level 2:** Moderate severity, some delays
- **Level 3:** Serious accidents, significant impact
- **Level 4:** Severe accidents, major disruptions



sample.csv



- Le
- Le

Data Preview

	State	City	Start_Lat	Start_Lng	Start_Time	Temperature(F)	Humidity(%)	Pressure(in)	Visibility(mi)
0	CA	Los Angeles	34.0522	-118.2437	2025-06-19 12:00:00	70	50	29.92	10
1	TX	Houston	29.7604	-95.3698	2025-06-19 14:30:00	75	60	29.8	8

Model

Pred

Severi

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- Le
- Le
- Le

Prediction Results

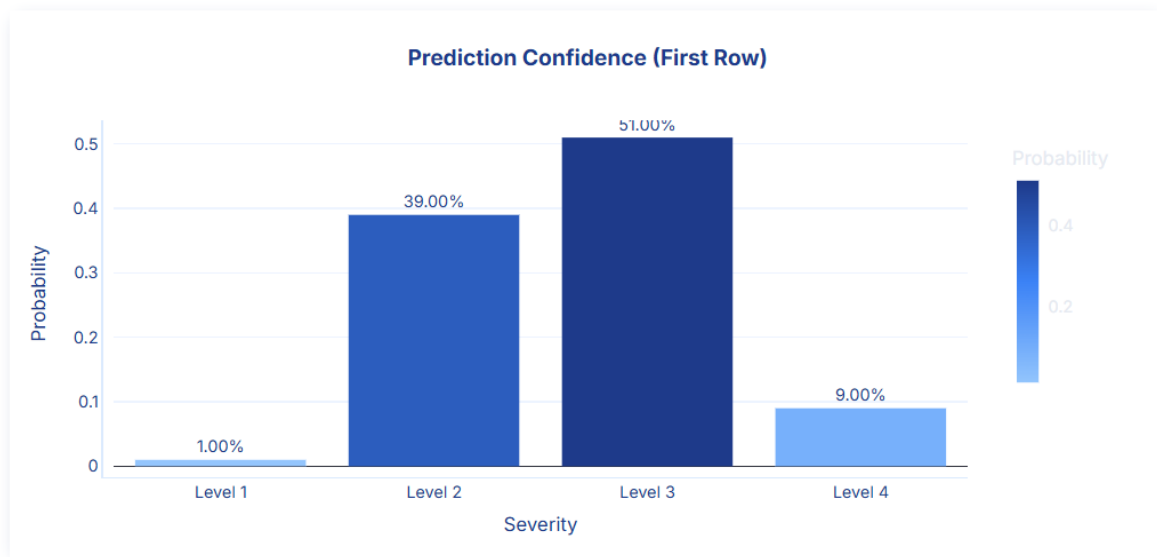
Successfully predicted 2 records!

The model has completed processing all records in your CSV file.

	State	City	Start_Time	Predicted_Severity	Severity_Label
0	CA	Los Angeles	2025-06-19 12:00:00	3	Serious
1	TX	Houston	2025-06-19 14:30:00	3	Serious

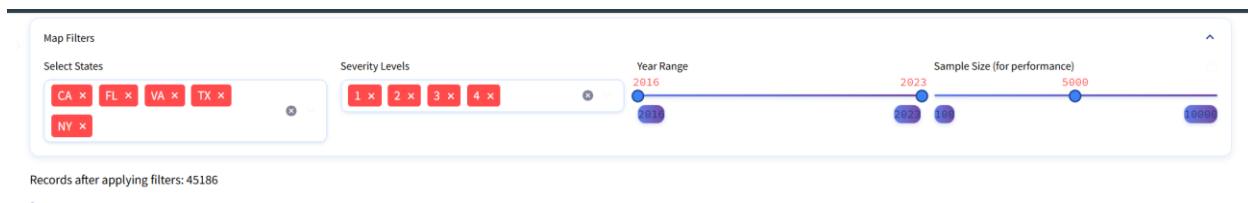
Download Predictions as CSV

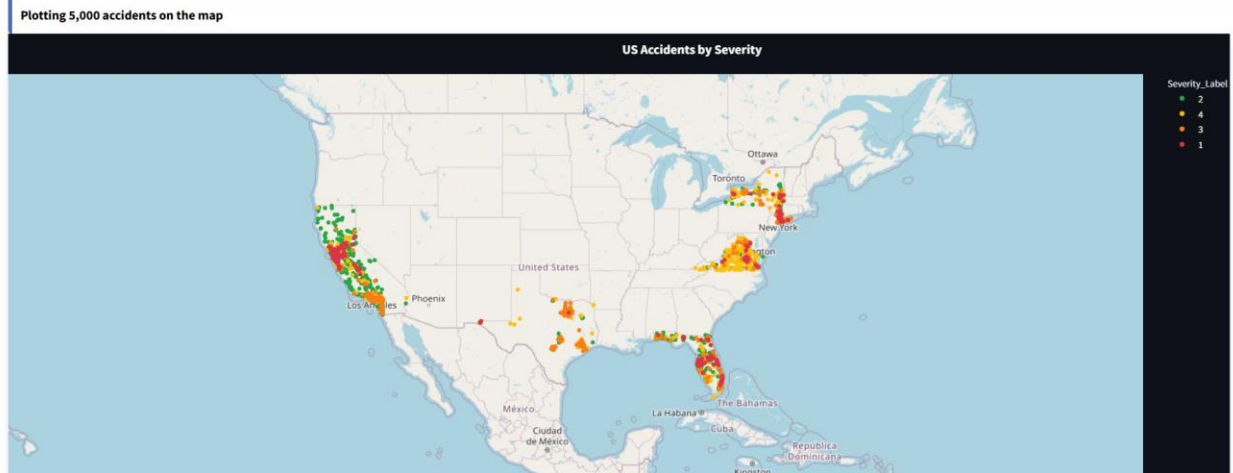
Sample Prediction Confidence (First Row)



- **User-friendly Input Form:** Select location, weather, time, and road conditions
- **Multiple Model Options:** Choose from PyCaret, AutoGluon, or baseline models
- **Real-time Predictions:** Instant severity level prediction
- **Prediction Confidence:** Probability scores for each severity level
- **Visual Results:** Charts showing prediction probabilities

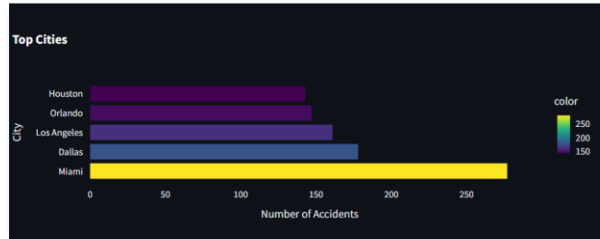
5. Geographic Analysis



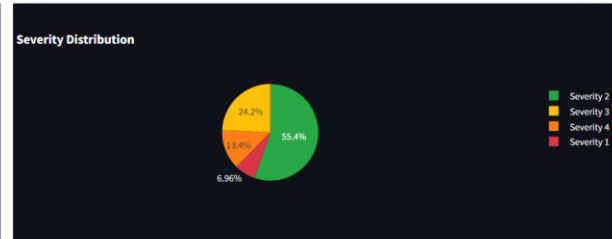


Map Statistics

Top 5 Cities by Accident Count



Severity Distribution



- Interactive map showing accident distributions
- State and city-level filtering
- Severity-based color coding
- Geographic clustering analysis

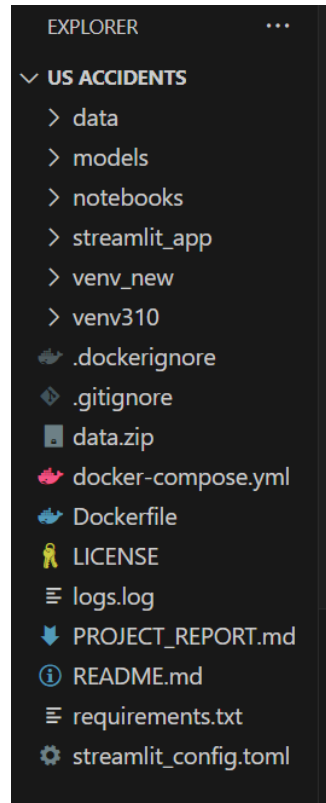
Technical Implementation

- **Framework:** Streamlit for rapid dashboard development
- **Visualization:** Plotly for interactive charts and maps
- **Styling:** Custom CSS for professional appearance
- **Configuration:** YAML-based configuration for easy customization
- **Model Integration:** Direct integration with trained ML models

User Experience Features

- **Responsive Design:** Optimized for different screen sizes
- **Interactive Elements:** Hover effects, clickable charts, dynamic updates
- **Professional Styling:** Clean, modern interface design
- **Real-time Updates:** Instant chart updates based on user selections

Project Structure



Deployment & Installation

Local Development Setup

Prerequisites

- Python 3.9+
- pip package manager
- 8GB+ RAM (recommended for large datasets)

Installation Steps

1. Clone the repository

```
git clone shazmin67/us-accidents
cd us-accidents-intelligence-hub
```

2. Create virtual environment

```
python -m venv venv310
source venv310/bin/activate # Linux/Mac
# or
venv310\Scripts\activate   # Windows
```

3. Install dependencies


```
pip install -r requirements.txt
```

```
# 4. Configure data paths
```

```
# Edit streamlit_app/config.yaml with correct paths
```

```
# 5. Run the application
```

```
cd streamlit_app
```

```
streamlit run app.py
```

Docker Deployment

Quick Start with Docker

```
# Build the container
```

```
docker build -t us-accidents-hub .
```

```
# Run the application
```

```
docker run -p 8501:8501 us-accidents-hub
```

Docker Compose (Production)

```
# Start all services
```

```
docker-compose up -d
```

```
# View Logs
```

```
docker-compose logs -f
```

```
# Stop services
```

```
docker-compose down
```

Production Deployment

- **Cloud Platforms:** AWS, GCP, Azure compatible
- **Container Orchestration:** Kubernetes ready
- **CI/CD:** GitHub Actions integration ready
- **Monitoring:** Health check endpoints included

Performance Metrics & Results

Model Performance Summary

Best Performing Model: PyCaret Extra Trees

- **Accuracy:** 90.6%
- **Use Case:** Production-ready severity prediction
- **Strengths:** High accuracy, robust to overfitting
- **Training Time:** Optimized through PyCaret automation

AutoGluon Ensemble Results

- **Accuracy:** 77.6%
- **Ensemble Size:** Multi-model stacking

- **Training Approach:** Automated hyperparameter tuning
- **Deployment:** Ready for real-time predictions

Sampling Strategy Impact

- **Original Data:** Higher overall accuracy (80.5%)
- **SMOTE Enhanced:** Better minority class handling (F1-macro: 0.391)
- **Class Balance Trade-off:** Accuracy vs. fairness consideration

Data Processing Efficiency

- **Original Dataset:** 7.7M records processed successfully
- **Feature Reduction:** 67 → 13 features (80% reduction)
- **Processing Pipeline:** Automated and reproducible
- **Memory Optimization:** Efficient handling of large datasets

Key Insights & Research Findings

Geographic Patterns

- **High-Risk States:** California, Texas, and Florida show highest accident concentrations
- **Urban vs Rural:** Urban areas demonstrate higher accident density but varied severity patterns
- **Interstate Analysis:** Highway accidents show distinct severity patterns compared to city roads

Temporal Patterns

- **Rush Hour Impact:** Clear peaks during 7-9 AM and 4-6 PM correlate with traffic patterns
- **Day-of-Week Effects:** Weekdays show higher frequency, weekends show different severity distribution
- **Seasonal Variations:** Winter months demonstrate higher average severity due to weather conditions

Weather Impact on Severity

- **Clear Weather:** Most frequent condition but shows varied severity outcomes
- **Adverse Weather:** Rain, snow, and fog correlate with higher severity levels
- **Visibility Factor:** Low visibility conditions significantly increase severe accident probability
- **Temperature Effects:** Extreme temperatures (very hot/cold) correlate with increased severity

Challenges Identified

1. Class Imbalance Challenge

The most significant challenge encountered was the severe class imbalance in the target variable:
 - **Severity 1:** Only 7.7% representation - **Severity 2:** Dominant at 43.3% - **Impact:** Traditional

models showed bias toward majority classes - **Solution Approaches:** SMOTE, undersampling, and balanced dataset creation

2. Feature Engineering Complexity

- **Temporal Features:** Extracting meaningful time-based patterns
- **Geographic Clustering:** Creating regional categories for better prediction
- **Weather Categorization:** Simplifying complex weather conditions

3. Model Selection Trade-offs

- **Accuracy vs Fairness:** High accuracy models often poor at minority class prediction
- **Computational Efficiency:** Ensemble models vs single model performance
- **Interpretability:** AutoML performance vs model explainability

Feature Importance Analysis

[Placeholder for Feature Importance Chart]

Top Predictive Features: 1. **Weather Conditions:** Primary factor influencing severity 2. **Time of Day:** Rush hour and night-time patterns 3. **Geographic Location:** State and urban/rural classification 4. **Road Infrastructure:** Traffic signals, junctions, crossings 5. **Environmental Factors:** Temperature, visibility, humidity

Technical Documentation & Libraries

1. **PyCaret:** Automated Machine Learning Library - <https://pycaret.org/>
2. **AutoGluon:** Amazon's AutoML Framework - <https://auto.gluon.ai/>
3. **Streamlit:** Web App Framework - <https://docs.streamlit.io/>
4. **Scikit-learn:** Machine Learning Library - <https://scikit-learn.org/>
5. **SMOTE:** Imbalanced-learn Library - <https://imbalanced-learn.org/>

Conclusion

This academic project successfully demonstrates the application of machine learning techniques to predict traffic accident severity using a large-scale, real-world dataset. The project addressed several key challenges in data science and machine learning:

Key Accomplishments:

1. **Handling Imbalanced Data:** Successfully implemented multiple strategies (SMOTE, undersampling) to address severe class imbalance in accident severity distribution.
2. **Model Performance:** Achieved 90.6% accuracy with PyCaret Extra Trees model, demonstrating the effectiveness of AutoML approaches for this problem domain.
3. **Comprehensive Analysis:** Conducted thorough exploratory data analysis revealing important patterns in geographic, temporal, and environmental factors affecting accident severity.

4. **Interactive Visualization:** Developed a professional dashboard for data exploration and model interaction, making complex analysis accessible and interpretable.
5. **Technical Proficiency:** Demonstrated competency in multiple ML frameworks (traditional ML, PyCaret, AutoGluon) and their comparative evaluation.

Academic Learning Outcomes:

- **Big Data Handling:** Experience processing and analyzing 7.7M+ records
- **Feature Engineering:** Creating meaningful features from complex temporal and geographic data
- **Model Evaluation:** Understanding trade-offs between accuracy and fairness in imbalanced datasets
- **Visualization Skills:** Creating effective data visualizations and interactive dashboards
- **Research Methodology:** Systematic approach to problem-solving and evaluation

Challenges and Limitations:

- **Class Imbalance:** Severity 1 accidents remain difficult to predict due to severe underrepresentation
- **Data Quality:** Some missing values and inconsistencies in the original dataset
- **Geographic Bias:** Dataset may be biased toward certain states with better accident reporting systems
- **Temporal Scope:** Limited to historical data without real-time prediction capabilities

This report represents the comprehensive documentation of the US Accidents Severity Prediction project, demonstrating the application of big data analytics and machine learning techniques for academic research purposes.